Question

Reacting $ZrCl_4$ with excess lithium borohydride in ether followed by vacuum distillation yields $Zr(BH_4)_4$. Treating the obtained $Zr(BH_4)_4$ with excess trimethylphosphine in ether and crystallizing with pentane produces an amber-colored solid. The byproduct of this reaction is a 1:1 Lewis acid-base adduct. X-ray diffraction results indicate that the amber-colored crystal is a dinuclear complex centered around zirconium atoms. Both zirconium atoms have an oxidation state of +4, with coordination numbers of 10 and 12, respectively. The ligands coordinated to zirconium are exclusively H^-, BH_4^-, and $P(CH_3)_3$. There are a total of three H^- ligands, all serving as edge-bridging groups connecting the two zirconium atoms (with no other edge-bridging ligands). Additionally, the mass fraction of zirconium in this amber-colored solid was measured to be 44.3%. Based on the above information, which of the following options is incorrect?

A. The ratio of the number of P atoms to B atoms in this compound is 2:5.

B. In this compound, the BH_4^- adopts two different coordination modes, with a ratio of 2:3 between the different coordination modes of this ligand.

C. All $P(CH_3)_3$ ligands are coordinated to the zirconium atom with a coordination number of 10.

D. The mass fraction of phosphorus in the by-products obtained from the preparation is 34.5%.

Answer

Correct Answer: B

Detailed Explanation

Based on charge balance, it can be deduced that the complex molecule contains a total of 3 H^- and 5 BH_4^-.

CHECKPOINT 0.5 PTS

The complex molecule contains a total of 3 H^- and 5 BH_4^-

The molar mass of the complex is $M = 91.22{\times}2{\div}0.4430 = 411.8(g/mol)$.

The number of $P(CH_3)_3$ units in the complex is:

$$[411.8 - 3{\times}1.008 - 5{\times}(4{\times}1.008 + 10.81) - 91.22{\times}2]{\div}(30.97 + 12.01{\times}3 + 1.008{\times}9) = 2$$

This indicates that the complex contains 2 $P(CH_3)_3$ ligands.

CHECKPOINT 0.5 PTS

The complex contains 2 $P(CH_3)_3$ ligands

Therefore, the chemical formula of the compound is $H_3Zr_2(P(CH_3)_3)_2(BH_4)_5$, and A is correct.

CHECKPOINT 1 PTS

The chemical formula of the compound is $H_3Zr_2(P(CH_3)_3)_2(BH_4)_5$

From this, the chemical reaction equation can be written as:

$$2Zr(BH_4)_4 + 5P(CH_3)_3 = H_3Zr_2(P(CH_3)_3)_2(BH_4)_5 + 3P(CH_3)_3BH_3$$

CHECKPOINT 1 PTS

The reaction equation is $2Zr(BH_4)_4 + 5P(CH_3)_3 = H_3Zr_2(P(CH_3)_3)_2(BH_4)_5 + 3P(CH_3)_3BH_3$

The mass percentage of phosphorus in $P(CH_3)_3BH_3$ is 34.5%, so D is correct.

CHECKPOINT 0.5 PTS

The mass percentage of phosphorus in $P(CH_3)_3BH_3$ is 34.5%

The total coordination number is 12+10=22. Typically, hydrogen in the complex acts as a bridging ligand, with the 3 H atoms providing a total of 6 coordination numbers. Each $P(CH_3)_3$ provides 1 coordination number, so among the remaining 5 BH_4^-, 4 act as tridentate ligands and 1 as a bidentate ligand.

CHECKPOINT 1 PTS

Among the 5 BH_4^-, 4 are tridentate ligands and 1 is a bidentate ligand

For the 12-coordinate zirconium atom, in addition to the three hydrogen atoms participating in coordination, there are 9 remaining coordination numbers. Considering steric effects, if $P(CH_3)_3$ ligands are present, their monodentate nature would cause excessive steric repulsion around the central Zr. Therefore, all its ligands are BH_4^-, with each BH_4^- acting as a tridentate ligand, totaling 3 such ligands.

CHECKPOINT 1 PTS

The 12-coordinate zirconium atom has 3 BH_4^- ligands, each acting as a tridentate ligand

For the 10-coordinate zirconium atom, it is surrounded by 2 $P(CH_3)_3$ ligands and the remaining 2 BH_4^- ligands.

The coordination number provided by BH_4^- is the total coordination number minus those from other ligands, i.e., $10-3-1\times2=5$. Thus, these two BH_4^- must consist of one tridentate and one bidentate ligand.

CHECKPOINT 1 PTS

The two BH_4^- ligands around the 10-coordinate zirconium atom consist of one tridentate and one bidentate ligand.

B is incorrect.

If both zirconium atoms had $P(CH_3)_3$ ligands, the steric hindrance for the 12-coordinate zirconium atom would be too large, ruling out this possibility. The two phosphine ligands must both coordinate to the 10-coordinate zirconium atom, so C is correct.

CHECKPOINT 1 PTS

Both phosphine ligands coordinate to the 10-coordinate zirconium atom